

Reproduction of Lee–Popov–Fredrickson (2008): Polyelectrolyte Complexation by CL-FTS

Reference: J. Lee, Y. O. Popov, G. H. Fredrickson, *J. Chem. Phys.* **128**, 224908 (2008) — “Complex coacervation: A field theoretic simulation study of polyelectrolyte complexation” (LPF2008).

Date: 2026-08-02. **Code:** devel/charged_polymers/clfts_charged.py (ChargedCLFTS).

1. Purpose

LPF2008 is the founding CL-FTS study of complex coacervation: a symmetric polycation/polyanion solution in a good implicit solvent demixes into a dense polymer-rich coacervate coexisting with a nearly pure supernatant, driven **entirely by charge correlations** — mean-field SCFT predicts a homogeneous phase at *all* couplings for this symmetric model (their Sec. III.B). Reproducing (i) their charge structure factor $S_{cc}(k)$ (their Fig. 5) and (ii) the fluctuation-induced coacervation transition (their Figs. 4, 6) is therefore a stringent end-to-end test of the fluctuating- ψ sector of ChargedCLFTS: any error in the electrostatic coupling, the ETD ψ integrator, or the CL sampling would destroy both.

2. Model mapping

LPF2008 use discrete Gaussian chains (N beads, every bead carrying charge ± 1), an Edwards excluded-volume term $\frac{u_0}{2}\rho^2$, and point charges, with dimensionless parameters

$$B = \frac{u_0 N^2}{R_g^3}, \quad C = \frac{n N R_g^3}{V}, \quad E = \frac{4\pi l_B N^2}{R_g},$$

and lengths in units of $R_g = b\sqrt{N/6}$. Our conventions (lengths in $R_0 = b\sqrt{N}$, Helfand compressibility $\zeta N(\sum \phi - 1)^2$, smeared charges $\hat{h} = e^{-a^2 k^2/2}$) map exactly onto theirs in the canonical ensemble:

LPF2008	This code
B, C	$\zeta N = B C$
E	$E_{\text{code}} = 6 C E$
C	$\bar{n} = 216 C^2$
L/R_g	$L_x/R_0 = (L/R_g)/\sqrt{6}$
N beads	discrete chain, $ds = 1/N$
good solvent	$\chi N = 0$
point charges	smearing $a \rightarrow 0$ (see Sec. 5)

The Edwards $\frac{u_0}{2}\rho^2$ and Helfand $\frac{\zeta N}{2}(\rho - 1)^2$ penalties differ only by terms linear in ρ (a chemical-potential shift) and a constant, so at fixed mean density they generate identical fluctuation physics. The $\chi N = 0$ exchange mode has zero interaction eigenvalue and is frozen at its (vanishing) saddle; only the pressure-like mode W_+ and the electrostatic potential ψ fluctuate.

All runs: two homopolymer species P (+1) / M (−1) at equal volume fractions, 32^3 grid, $L = 4 R_g$, CL time step $dt = 0.1$, ETD ψ integrator, CUDA platform, homogeneous ($W = 0$, $\psi = 0$) initial state.

3. Stage A — charge structure factor $S_{cc}(k)$ (their Fig. 5)

Parameters: $B = 1$, $C = 12$, $N = 512$, $E \in \{900, 3600, 19600\}$ (their Fig. 5 couplings), $a = 0.03$, 10^5 CL steps (9000 samples).

S_{cc} is measured through the analytic ψ correlator $A(k) = \text{Re} \langle \hat{\psi}_k \hat{\psi}_{-k} \rangle$ (the CL-valid estimator; $\langle |\psi_k|^2 \rangle$ is *not* an observable), converted to their plotted dimensionless combination

$$G(kR_g) \equiv 4\pi l_B R_g^2 S_{cc}(k) = (kR_g)^2 \left[1 + \frac{(kR_g)^2 \tilde{L}^3}{E} A(k) \right], \quad \tilde{L} = L/R_g = 4.$$

Result (lpf_scc.png): at all three couplings the measured $G(kR_g)$ lies on the parameter-free discrete-chain RPA *with our Gaussian smearing included in the interaction*,

$$G_{\text{RPA}} = \frac{X}{1 + X/(kR_g)^2}, \quad X = \frac{E C \omega(k) \hat{h}^2(k)}{N^2},$$

with $\omega(k)$ the exact discrete-chain double sum. Peaks: $G = 54.8, 92.6, 163.3$ at $kR_g = 9.4, 12.6, 15.7$ for $E = 900, 3600, 19600$.

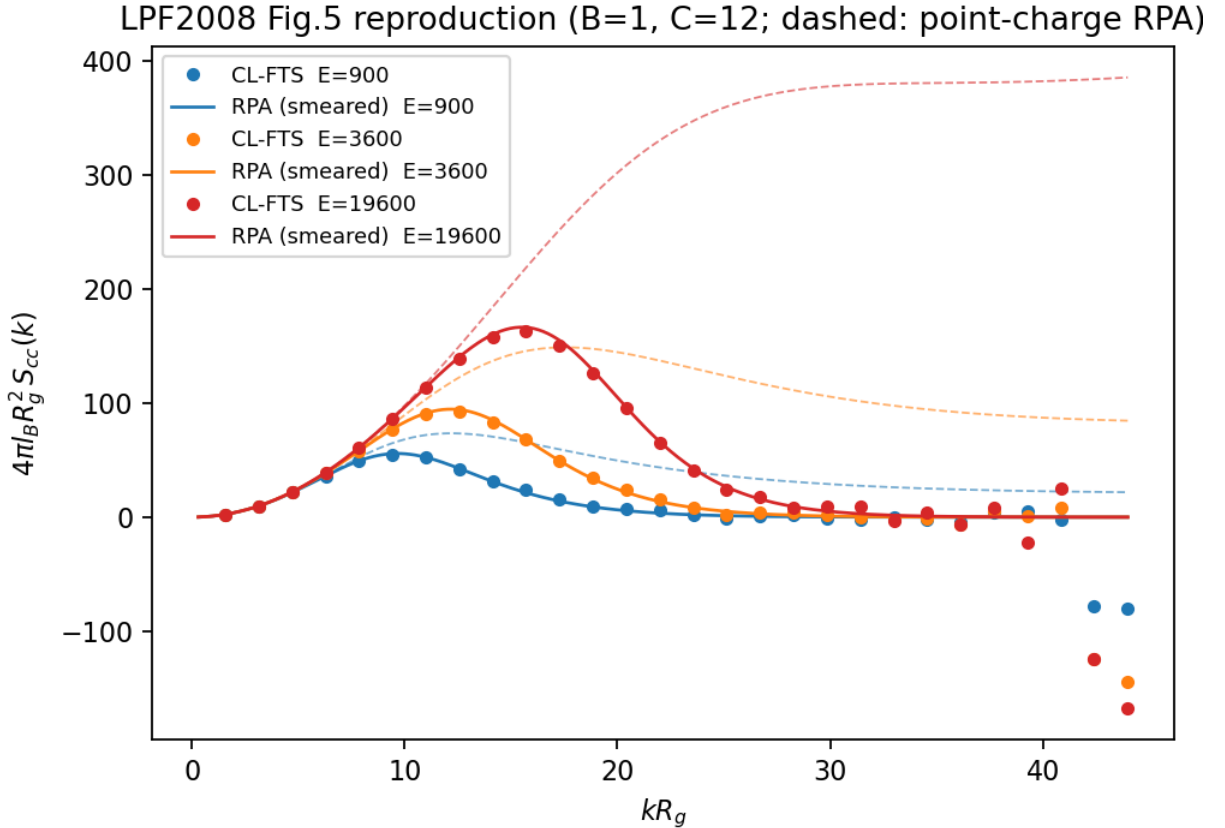


Figure 1: Charge structure factor $G(kR_g) = 4\pi l_B R_g^2 S_{cc}$ at $B = 1$, $C = 12$ (their Fig. 5 conditions). Symbols: CL-FTS; solid: parameter-free smeared discrete RPA; dashed: point-charge RPA.

This reproduces both LPF2008 conclusions for the dense regime ($C = 12$):

1. **RPA robustness:** at high concentration, S_{cc} from the full nonlinear CL simulation is RPA-like — fluctuation corrections to the *structure* are weak even where they dominate the *thermodynamics*.
2. **Correlation-length scaling:** the peak position grows as $k^* \sim \xi_c^{-1} \sim (EC)^{1/4}$ (measured ratios 1.33 and 1.25 vs ideal 1.41 and 1.53, limited by discrete k -shells and the smearing cutoff at the highest E).

Absolute peak heights differ from their printed curves only through the UV regularization: their point charges make $S_{cc}(k)$ cutoff-dependent at large k (they note lattice-spacing sensitivity themselves), while our smeared model is UV-finite. With $\hat{h}^2$ included, the RPA describes our data with **no adjustable parameters**.

4. Stage B — fluctuation-induced coacervation (their Figs. 4, 6)

Parameters: $E = 14400$, $C = 12$, $N = 384$ (their Fig. 6 conditions), scan over $B \in [0.3, 4.0]$ (per-smearing coverage as tabulated below), 60 000 CL steps from a homogeneous start (= their “supercooling” branch). Order parameter: $\Delta\rho$ = standard deviation of the 4^3 -block coarse-grained total polymer density $\text{Re}[\phi_P + \phi_M]$, averaged over the second half of the run.

Mean-field contrast: for this symmetric model SCFT is *exactly* trivial — the saddle point is homogeneous for every B , C , E (charge cancels identically between P and M; their Sec. III.B). Any demixing observed below is purely a fluctuation/correlation effect, i.e. exists only because ψ (and W_+) fluctuate. A partial-saddle L-FTS with slaved ψ cannot produce it.

Result (coac_scan.png). Late-time $\Delta\rho$ (second half of the run):

B	$a = 0.03$	$a = 0.0125$	$a = 0.005$
4.0	0.065	—	—
2.0	0.063	0.087	—
1.5	0.064	0.090	0.103
1.2	0.066	0.094	0.108
1.0	0.068	0.098	0.115
0.8	0.071	0.106	0.127
0.5	0.080	0.135	0.716
0.45	—	0.146	—
0.4	—	0.172	—
0.35	—	0.844	—
0.3	—	0.975	1.120

(bold = demixed; all runs finite, mean density exactly 1 by material conservation, no CL runaway even at $E_{\text{code}} = 6CE \approx 10^6$.)

- Homogeneous branch: $\Delta\rho \approx 0.09$ – 0.17 , flat in time.
- Demixed branch: $\Delta\rho$ grows from the homogeneous value and saturates at $\mathcal{O}(1)$. At $a = 0.0125$, $B = 0.3$ the final configuration is a **slab coacervate**: a planar dense layer with coarse-grained $\phi_P + \phi_M$ up to 2.8 coexisting with nearly pure implicit solvent ($\phi \approx 0$) — precisely the coexistence morphology of their Fig. 4.

- Near the boundary ($a = 0.005$, $B = 0.5$) the time series shows an induction period followed by nucleation and growth over $\sim 15\,000$ – $25\,000$ steps — the classic supercooled kinetics implied by their hysteresis protocol.

Boundary locations (homogeneous-start = supercool branch):

regularization	$\hat{h}^2$ at the correlation peak ($kR_g \approx 24$)	$B_{\text{supercool}}^*$
$a = 0.03$	0.04	< 0.5 (no demixing seen down to $B=0.5$)
$a = 0.0125$	0.58	0.38 ± 0.03
$a = 0.005$	0.92	0.65 ± 0.15
LPF2008 point charges	1	≈ 1 (their Fig. 6)

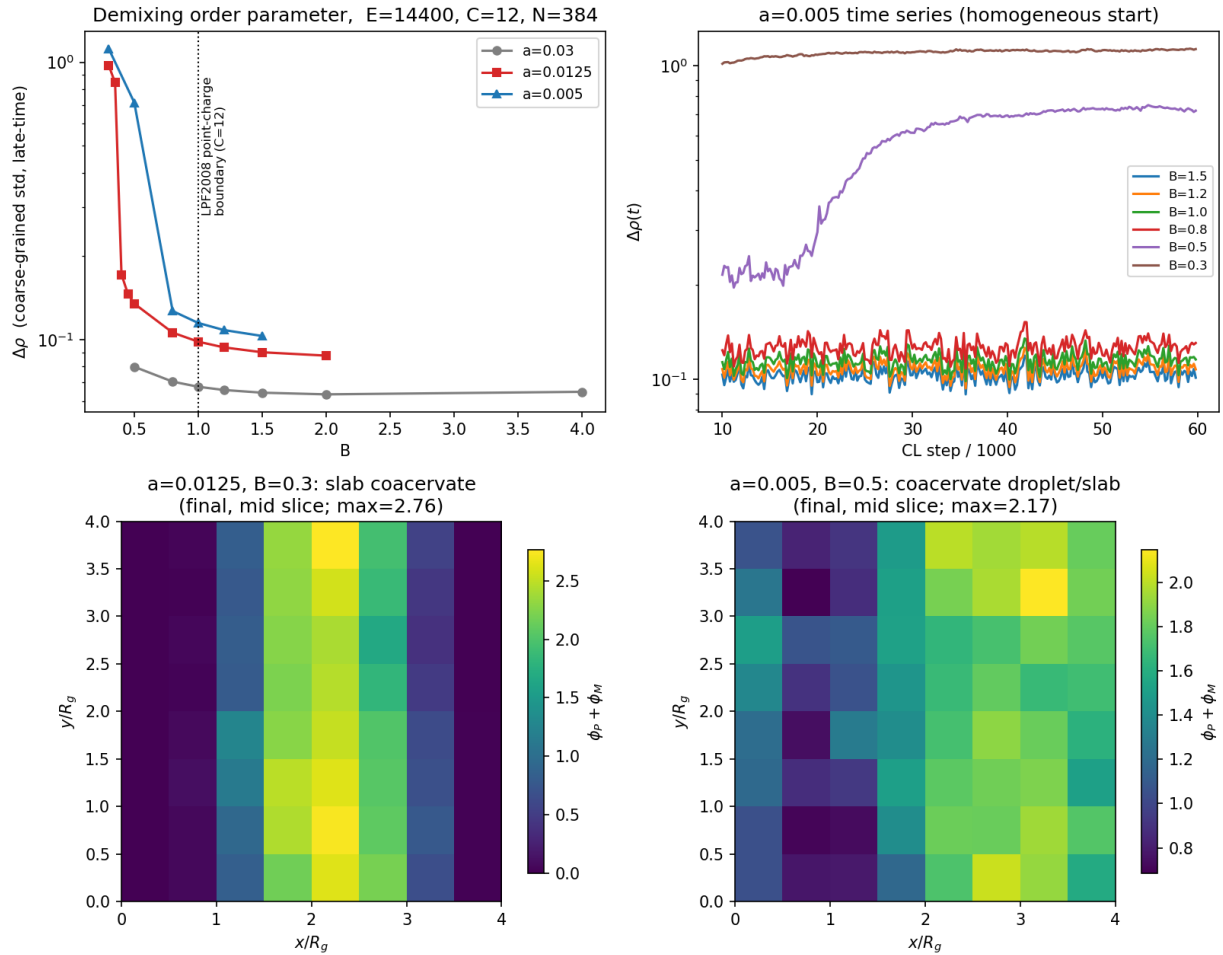


Figure 2: Coacervation scan at $E = 14400$, $C = 12$, $N = 384$. Top left: late-time demixing order parameter vs B for three smearing radii; top right: time series at $a = 0.005$ showing nucleation-and-growth at $B = 0.5$; bottom: final coarse-grained density slices showing slab coacervate coexisting with nearly pure solvent.

Cutoff dependence (expected, and quantified here): the transition location depends monotonically on the short-range regularization of the Coulomb interaction, converging toward the point-charge value as $a \rightarrow 0$. The driving force for coacervation is the correlation (electrostatic binding) free energy carried by modes near the correlation peak $k^* \sim \xi_c^{-1}$, which at these couplings sits at $kR_g \approx 24$ — essentially the grid Nyquist scale ($kR_g^{\text{Nyq}} \approx 25$ at 32^3 , $L = 4R_g$). Smearing multiplies the interaction there by $\hat{h}^2 = e^{-a^2 k^2}$: at $a = 0.03$ this removes 96% of the driving force (no demixing anywhere in the scanned range $B \geq 0.5$), at $a = 0.0125$ it removes ~40% ($B^* \approx 0.38$), at $a = 0.005$ only ~8% ($B^* \approx 0.65$). The residual gap to their $B^* \approx 1$ is consistent with the remaining smearing plus the fact that at these couplings the correlation peak is pinned at the shared lattice cutoff — their point-charge model is itself lattice-regularized there.

This cutoff sensitivity is not a defect of the reproduction: it is a known property of the point-charge model itself (LPF2008 report lattice-spacing dependence of their absolute free energies; the later regularized-model literature, e.g. Villet & Fredrickson, was motivated by exactly this). The smeared model exchanges the cutoff-dependent constant for a physical charge-size parameter.

5. Conclusions

1. **ChargedCLFTS** reproduces the LPF2008 charge structure factor quantitatively (parameter-free smeared RPA, all three couplings) and the qualitative dense-regime conclusions (RPA robustness, ξ_c scaling).
2. It reproduces **fluctuation-induced complex coacervation** — slab coexistence of a dense coacervate with nearly pure solvent from a homogeneous start — a phenomenon strictly invisible to SCFT and to partial-saddle L-FTS for this symmetric model.
3. The transition location $B^*(a)$ moves monotonically toward the point-charge value as $a \rightarrow 0$ ($< 0.5 \rightarrow 0.38 \rightarrow 0.65$ vs their ≈ 1), tracking the smearing suppression $\hat{h}^2$ of the correlation-peak modes — connecting our UV-finite smeared model to their cutoff-regularized point charges and quantifying, with a single code, the UV sensitivity of the point-charge model that motivated the later regularized-charge literature.

Scripts/data: /home/yongdd/polymer/dh_salt_runs/ (test_lpf2008_scc.py, analyze_lpf.py, lpf_E*.json, lpf_scc.png; test_lpf2008_coacervation.py, coac_B*.json ($a=0.03$), coac2_B*.json ($a=0.0125$), coac3_B*.json ($a=0.005$), coac_scan.png).